

TrES-5b

R-1999

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_150911 · created 2026-08-07T15:52:08+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457276.7951 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.185 +/- 0.053
Transit depth (Rp/Rs)^2	3.41 +/- 1.98 [%]
Orbital Inclination (inc)	89.24 +/- 2.91
Airmass coefficient 1 (a1)	0.21 +/- 0.1
Airmass coefficient 2 (a2)	1.03 +/- 0.49
Scatter in the residuals of the lightcurve fit is	42.9 %
Best Comparison Star	#3 - [174, 36]
Optimal Aperture	1.93
Optimal Annulus	4.75
Transit Duration (day)	0.0871 +/- 0.0085

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.185 $\pm$ 0.053 vs 0.142 (deviation 0.66 σ)
Duration fit vs published	0.0871 d vs 0.0758 d (deviation 1.08 σ)
Mid-time O-C	4.0 min
Depth SNR	2.9
Residual scatter	42.9 %

Light curve

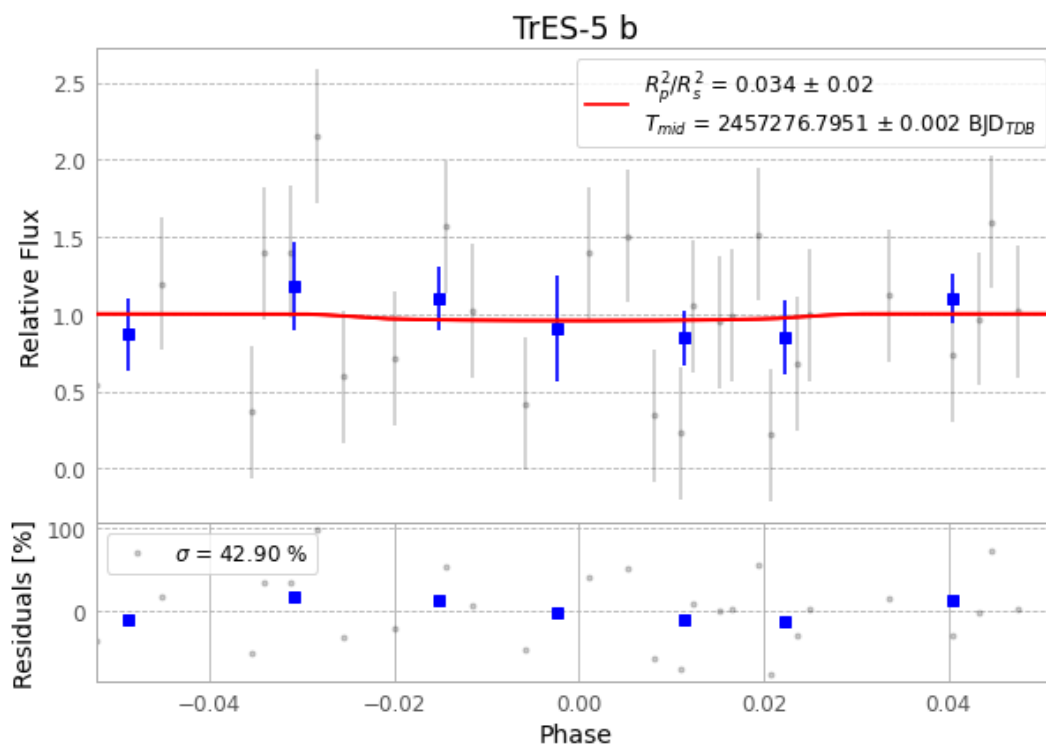


Figure 1 — FinalLightCurve_TrES-5 b_2015-09-10.png (SHA-256 8f59a5d0858b0e67...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [430, 290], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c62ff535ee58bcd9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

77 FITS frames (51 MB), TRES-5150911050312.FITS → TRES-5150911085112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-150911.log (721599b081be...), FinalLightCurve_TrES-5 b_2015-09-10.png (8f59a5d0858b...), FinalLightCurve_TrES-5 b_2015-09-10.csv (bf4bdc4e9009...)

Compute resources

Wall time 131.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 15:52Z.